SlideCanon Course Notes

From low quality to high quality

What the model learns to do (denoise + deblock + upscale)

Generated from structured markdown on 2026-04-24. This PDF keeps slide content, images, and speaker notes in a more readable A4 note format.

Contents

1. From low quality to high quality
2. Design goals (practical constraints)
3. Where I started
4. What is SOTA depends on the protocol
5. Why tile the image?
6. Training data
7. Network architecture (model B)
8. Evaluation
9. Takeaway

Slide 1

From low quality to high quality



In-Depth Explanation

This is the only “app context” slide: the input is a low-resolution, recompressed image and the goal is to restore a clean, sharp, high-resolution result.
In this talk I focus on how I designed the training setup and the network that powers this transformation, not on the app itself.

Slide 2

Design goals (practical constraints)

- Runs locally on older phones (latency + memory)
- Privacy by design (no upload)
- Better on people: faces, skin, hair detail
- Fewer hallucinations / no “stable-diffusion look”

In-Depth Explanation

Shortly: this is not a “largest model wins” project. The constraints are mobile compute and memory, and the requirement that processing stays on device for privacy.
Visually, the model should help the content users care about most—people—without inventing details.

Slide 3

Where I started

classic ESRGAN direction

- CNN SR → SRGAN/ESRGAN: sharpness via perceptual + adversarial loss
- Great for “wow” textures, risky for truthfulness
- Training sensitive to degradation realism
- For mobile: big models and instability are a problem

In-Depth Explanation

My starting point was the classic super□resolution evolution: first regression CNNs optimized for PSNR, then perceptual and GAN-based methods like SRGAN and ESRGAN.
The key advantage of ESRGAN-style training is that it can produce visually sharp outputs even when pixel-wise losses would look too smooth.
But that same property is a risk: adversarial and perceptual objectives can encourage “plausible” texture that is not necessarily the true detail from the original scene.
This becomes especially problematic for faces and for repeated patterns where hallucinations look like artifacts.
Another practical issue is stability: GAN training can be sensitive, and larger ESRGAN-like backbones can be too heavy for older phones.
So I treated ESRGAN as a reference point, but changed the approach significantly to fit my constraints.

Slide 4

What is SOTA depends on the protocol

- Benchmarks (PSNR/SSIM) favor fidelity-oriented models
- Real photos need blind restoration (unknown blur/noise/JPEG chains)
- Perceptual metrics help, but can reward hallucinations
- 2026 trend: hybrid pipelines (restoration backbone + generative prior)

In-Depth Explanation

When people ask “what is state of the art?”, the honest answer is: it depends on the evaluation protocol.
If you evaluate on synthetic benchmarks with known downsampling, fidelity-oriented models—often large transformers—can dominate PSNR and SSIM.
But real photos are different: you have unknown blur, sensor noise, sharpening, and almost always a compression and resizing chain.
In that setting, you need blind restoration that is robust to degradation mismatch.
Perceptual metrics like LPIPS are useful because they correlate better with human judgment than PSNR alone, but they can also reward outputs that look plausible rather than correct.
The direction around 2026 is increasingly hybrid: keep a strong restoration backbone, and optionally add a generative prior, but you must control it to avoid “AI look”.
That observation strongly shaped my design choices.

Slide 5

Why tile the image?

- Whole-image SR is memory-heavy at high resolution
- Process fixed-size tiles (e.g., 256×256) and stitch back



In-Depth Explanation

Main idea: instead of feeding a full 4K image into a huge network, we split the image into tiles.
Tiles have a fixed size, so memory use is stable and predictable on mobile.
We overlap tiles so that boundaries are not visible.
Then we stitch the final image by blending the overlaps with a smooth mask—this prevents seams while keeping details.

Slide 6

Training data

pairs from real photos

- High-quality crops are ground truth targets
- Low-quality inputs are synthesized (downscale + JPEG + nearest-up)



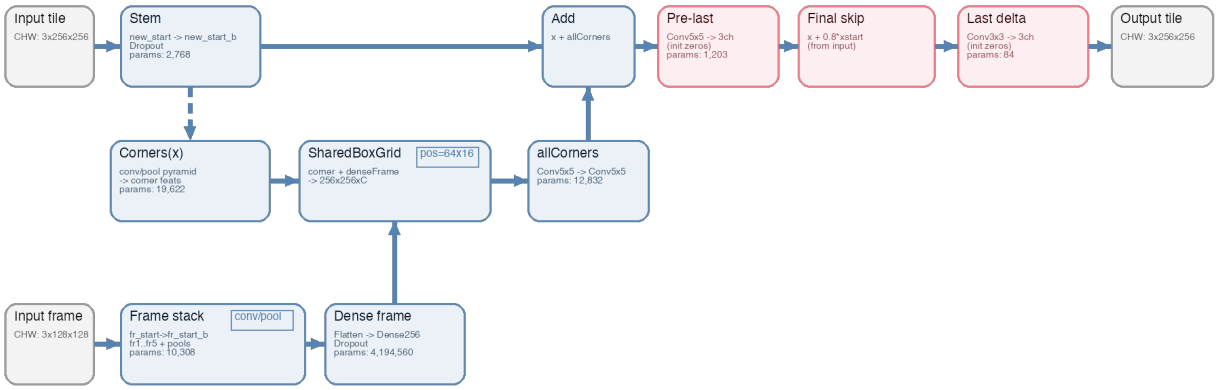
In-Depth Explanation

The training set is built from high-quality photos by sampling many 256×256 crops as ground truth.
For each crop we synthesize a degraded input by downscaling, compressing with JPEG, and upscaling back with nearest neighbor—this matches the type of artifacts seen in practice.
This gives an unlimited stream of paired examples without manual labels.
In addition, crop sampling can be biased toward regions that matter, like faces and high-detail areas, to improve quality where users notice it most.

Slide 7

Network architecture (model B)

Two inputs: tile + frame context, plus corner/grid injection



In-Depth Explanation

This diagram summarizes the model B architecture from the training notebook.
The key feature is combining the main tile stream with a lightweight context stream and an explicit corner/grid injection path.

Slide 8

Evaluation

what I report (brief)

- PSNR / SSIM: distortion (reference-based)
- LPIPS: perceptual similarity (reference-based)
- Visual checks: faces, text, repeated patterns
- Runtime + memory on-device (tile size matters)

In-Depth Explanation

I keep evaluation lightweight for the presentation: PSNR/SSIM for distortion, LPIPS for perceptual quality, plus targeted visual checks.
For this project, runtime and memory on a phone are first-class metrics.

Slide 9

Takeaway

Engineering the training pipeline mattered as much as the model

In-Depth Explanation

The main contribution is not a single trick, but a controlled pipeline: realistic degradations, deterministic crop selection, a small on-device architecture, and stitching via overlap + mask to avoid seams.